4.HIERARCHICAL INHERITANCE

```
#include <iostream>
using namespace std;

class ArrayInput {
protected:
   int arr[100];
   int size;
public:
   void getArray() {
      cout << "Enter size of array: ";
      cin >> size;
      cout << "Enter elements: ";
      for (int i = 0; i < size; i++) {
         cin >> arr[i];
      }
   }
   void displayArray() {
      cout << "Array elements: ";
      for (int i = 0; i < size; i++) {
         cout << arr[i] << " ";
      }
      cout << endl;
   }
};

class ArraySum : public ArrayInput {
public:
   void findSum() {
      int sum = 0;
      for (int i = 0; i < size; i++) {
         sum += arr[i];
```

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        }
        cout << "Sum of elements: " << sum << endl;
    }
};

class ArrayMax : public ArrayInput {
public:
    void findMax() {
        int maxVal = arr[0];
        for (int i = 1; i < size; i++) {
            if (arr[i] > maxVal) {
                maxVal = arr[i];
            }
        }
        cout << "Maximum element: " << maxVal << endl;
    }
};

int main() {
    ArraySum obj1;
    ArrayMax obj2;

    cout << "Enter array for sum calculation:\n";
    obj1.getArray();
    obj1.displayArray();
    obj1.findSum();

    cout << "\nEnter array for max calculation:\n";
    obj2.getArray();
    obj2.displayArray();
    obj2.findMax();
```

```
    return 0;
}
```

```
Enter array for sum calculation:
Enter size of array: 5
Enter elements: 1 2 3 4 5
Array elements: 1 2 3 4 5
Sum of elements: 15

Enter array for max calculation:
Enter size of array: 5
Enter elements: 2 3 5 6 1
Array elements: 2 3 5 6 1
Maximum element: 6

C:\Users\yoges\OneDrive\Documents>
```